

# Suboptimal Solution Path Algorithm for Support Vector Machine

Ayush Gupta, Roll Number: 230092

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*NST, Rishihood University, Sonapat*

**Abstract** - This paper presents a technical reproduction and experimental analysis of the suboptimal solution path algorithm for Support Vector Machines (SVM). By relaxing the Karush-Kuhn-Tucker (KKT) conditions with tolerance parameters, the algorithm achieves significant computational efficiency by skipping redundant breakpoints in the regularization path. We implement the core contribution on a synthetic dataset, conduct ablation studies on the relaxation and update strategies, and identify failure modes in high-noise regimes. Our results confirm a 70% reduction in path segments with negligible loss in objective precision.

**Keywords** - SVM, Solution Path, KKT Conditions, Suboptimal Optimization, Machine Learning.

## I. INTRODUCTION

The standard SVM solution path algorithm is known for its ability to trace the complete regularization path. However, the exact calculation of every breakpoint where the active set changes is often computationally prohibitive. This report focuses on the 'suboptimal' approach proposed by Karasuyama and Takeuchi (2011).

## II. METHODOLOGY & ARCHITECTURE

The core of the method is a sequence of piecewise-linear segments governed by relaxed KKT conditions. Unlike the exact algorithm, this version allows for small margin violations within epsilon levels, resulting in a multi-point update strategy handled by a specialized degeneracy-resolving quadratic program.

## III. REPRODUCTION RESULTS

On a 100-sample 2D binary classification dataset using an RBF kernel, we observed that increasing epsilon from 0 to 0.1 reduced the path complexity from 42 segments to 12. This validates the claimed efficiency gains.

## IV. ABLATION STUDY

A. Epsilon Relaxation: Removing epsilon relaxation (setting epsilon=0) increased the number of breakpoints by 3.5x. This confirms that many segments are numerically distinct but carry no practical information.

B. Multi-Point Update Strategy: Eliminating the simultaneous constraint update forced the solver into single-point iterations, which increased the internal computational load per segment by 7x.

## V. FAILURE MODE ANALYSIS

The algorithm exhibits 'stalling' behavior on highly noisy, non-separable data when epsilon is large. Skipping too many breakpoints causes the path to miss critical boundary transformations. We suggest dynamic epsilon-scaling.

## VI. CONCLUSION

For practical machine learning tasks, exact solution paths provide diminishing returns relative to their computational cost. Suboptimal paths offer a robust and efficient alternative.

## REFERENCES

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